

DECODELABS INTERNSHIP



PROJECT 1: DATA CLEANING & PREPARATION

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1. Introduction

Data cleaning is an important step in data analytics. Raw datasets often contain missing values, duplicate records, and inconsistent formats. The purpose of this project is to improve data quality and prepare the dataset for analysis.

2. Objective

The main objective of this project is to clean and prepare a raw dataset by:

- Identifying missing values
- Checking duplicate records
- Standardizing text values
- Formatting dates consistently
- Improving overall data quality

3. Tools Used

- Microsoft Excel
- Data Cleaning Techniques

4. Tasks Performed

4.1 Missing Values Handling

The dataset was checked for missing values. Missing age values were identified and filled appropriately to maintain data consistency.

4.2 Duplicate Record Verification

The dataset was examined for duplicate records using Excel's Remove Duplicates feature. Duplicate entries were verified and checked.

4.3 Data Standardization

Different city name formats were standardized.

Examples:

- Bengaluru → Bangalore
- bangalore → Bangalore
- Mysuru → Mysore

4.4 Date Format Standardization

All date values were converted into a consistent DD-MM-YYYY format.

5. Results

Activity	Status
Missing Values Checked	Completed
Missing Values Fixed	Completed
Duplicate Records Verified	Completed
City Names Standardized	Completed
Date Formats Standardized	Completed
Dataset Cleaned	Completed

6. Conclusion

The dataset was successfully cleaned and prepared for analysis. Missing values were handled, duplicate records were verified, city names were standardized, and date formats were corrected. The final dataset is accurate, consistent, and ready for further analytical tasks.

7. Outcome

This project helped in understanding the importance of data cleaning and preparation in real-world data analytics workflows. Proper data cleaning improves data quality and ensures reliable analysis results.